



**R.M.K. COLLEGE OF ENGINEERING AND TECHNOLOGY
DEPARTMENT OF ELECTRONICS AND COMMUNICATION
ENGINEERING**

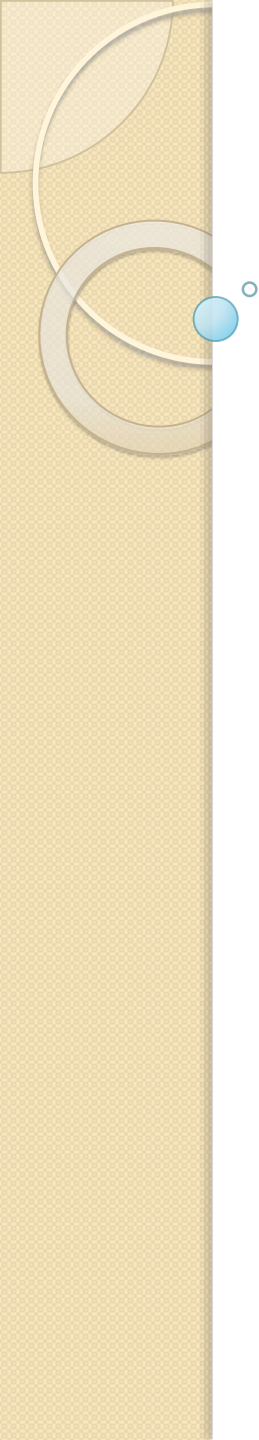
B.E – PRODUCT DEVELOPMENT LAB- First Review - DATE: 13.02.2026

ZONE-LEVEL REAL TIME ACCIDENT DETECTION SYSTEM

**SDG – 11 SUSTAINABLE CITIES AND
COMMUNITIES**

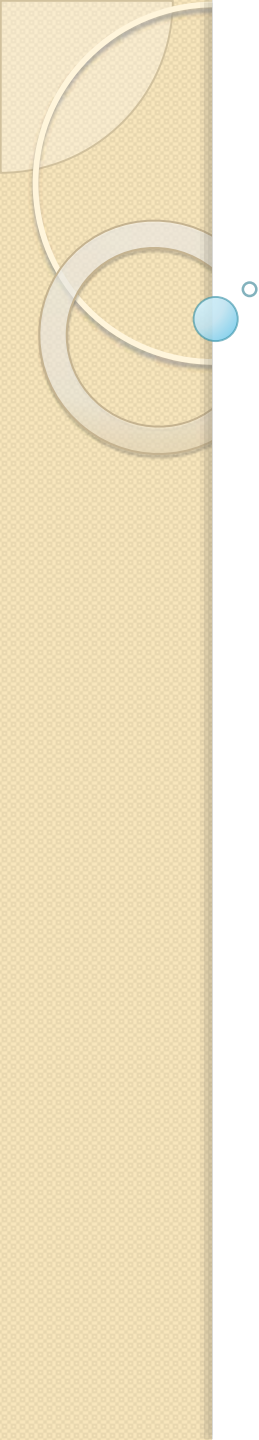
DHIVYASHREE C	-	111624104020
INDUJA M	-	111624104032
KEERTHI J	-	111624104035

Under the Guidance of
Dr.S.MUTHUMANICKAM
Professor
DEPARTMENT OF ECE



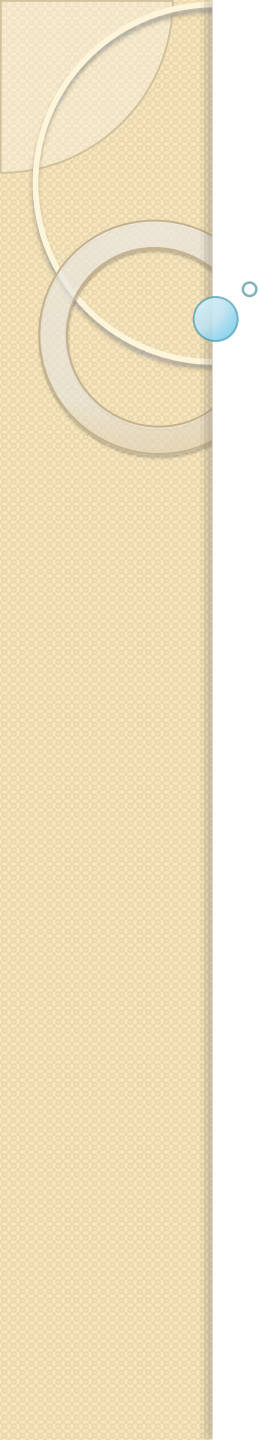
CONTENTS

- OBJECTIVE
- INTRODUCTION
- LITERATURE REVIEW
- PROBLEM STATEMENT
- PROPOSED WORK
- ALGORITHM
- FLOW DIAGRAM
- BLOCK DIAGRAM
- CONCLUSION
- REFERENCES



OBJECTIVES

- To develop an intelligent vehicle safety monitoring system.
- To detect accidents automatically based on sudden abnormal motion or impact.
- To ensure driver safety by verifying basic safety conditions before vehicle operation.
- To automatically identify and track the vehicle's location during emergencies.
- To transmit accident alerts to the nearest zone control station immediately.
- To reduce emergency response time and improve rescue efficiency.
- To minimize false accident alerts through a manual cancellation mechanism.
- To design a compact, cost-effective, and reliable real-time safety solution.

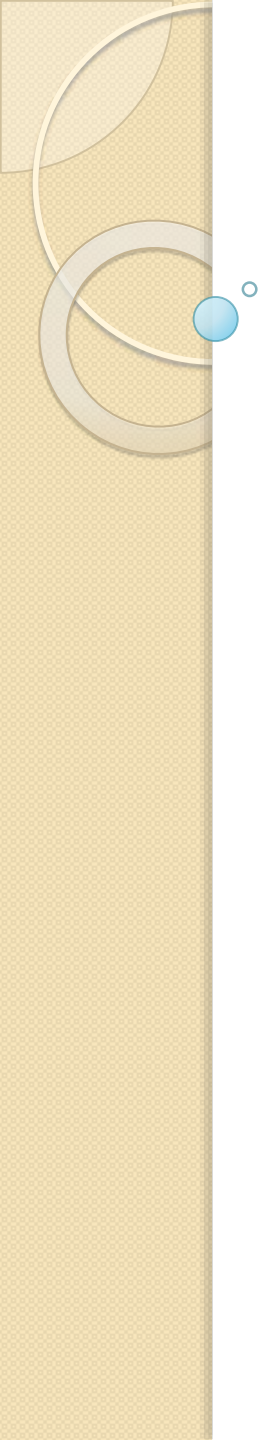


INTRODUCTION

- Road accidents cause serious injuries and loss of life due to delayed emergency response.
- Immediate accident detection and quick alert transmission can save lives.
- This project develops a Zone-Level Vehicle Accident Detection and Alert System.
- The system automatically detects accidents and sends alerts with location details.
- It aims to reduce response time and improve road safety through real-time monitoring.

Literature Review

SI. NO	AUTHOR/YEAR	METHODOLOGY	ALERT MECHANISM	LIMITATION	PROPOSED System Improvements
1	AcciRadar (2025) Janani Sindhu et al.	CCTV Surveillance Cameras, YOLOv8, CNN, Raspberry Pi, GPS, Twilio API	Automated voice call + SMS to nearest hospital with GPS location	Dependent on presence of surveillance cameras; requires internet connectivity	Uses RF-based tower alerts without relying on cameras or cloud connectivity
2	Collision Detection & Alert System (2025) Dr. Sumitra et al	Arduino, ADXL Accelerometer, GPS Neo6M, GSM 900A	SMS alert with GPS location to emergency contacts	GSM delays in low network areas; prone to false triggers	RF communication ensures alerts even in network-poor regions; STOP button avoids false alerts
3	Vehicle Accident Prevention & Detection via App (2025) M. Sumithra et al.	IR, Alcohol Sensor, Accelerometer, LiDAR, Radar, Arduino, GSM, Mobile App	Real-time SMS alerts via GSM and mobile notifications through app	Complex multi sensor setup; network dependency; mobile app required	Simplified microcontroller based design; offline zone level transmission; cost effective



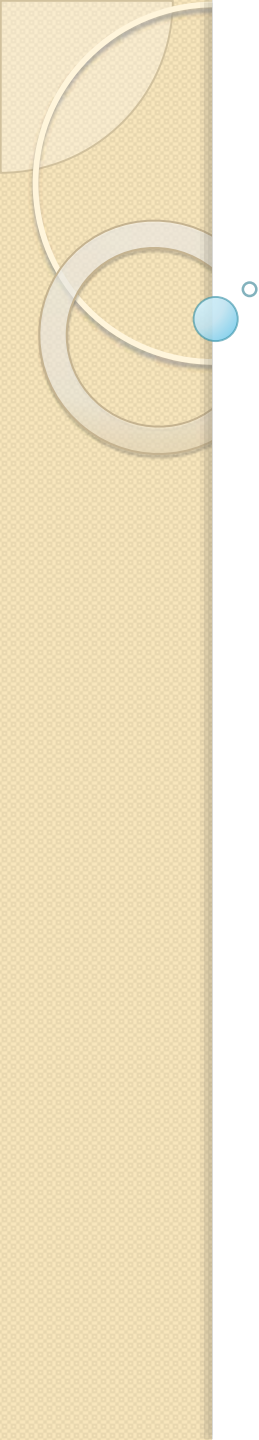
PROBLEM STATEMENT

Manual reporting of accidents depends on witnesses and mobile communication.

In remote areas:

- victims may be unconscious
- no one available to inform
- rescue is delayed

Therefore, a system is required to automatically detect accidents and inform nearby authorities immediately.



PROPOSED WORK

- The system detects accidents using an accelerometer and verifies driver condition using a response button.
- If the driver does not respond, the system sends GPS coordinates wirelessly to a zone monitoring station.
- The operator receives the alert and informs emergency services.

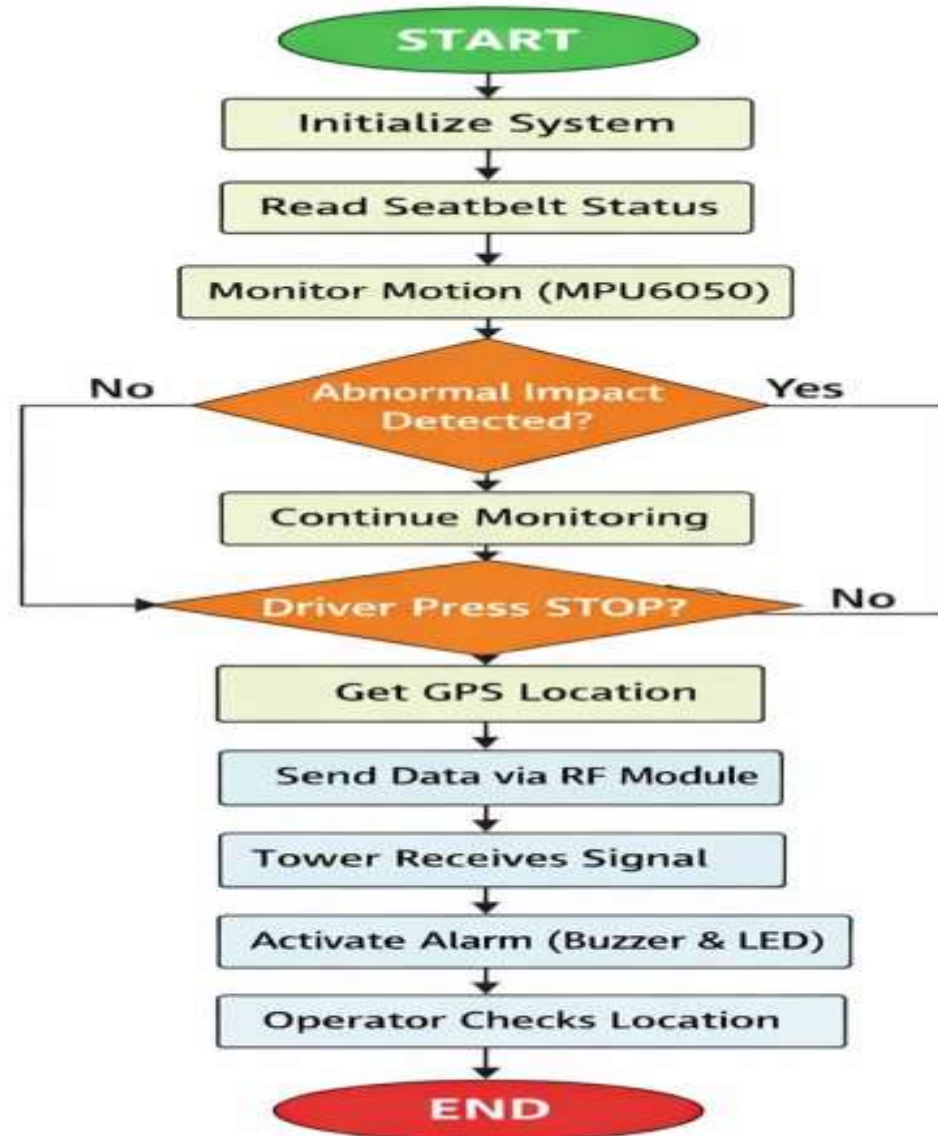
Main features:

- ✓ Automatic crash detection
- ✓ Driver confirmation using voice alert
- ✓ Wireless zone-level communication
- ✓ Faster rescue support

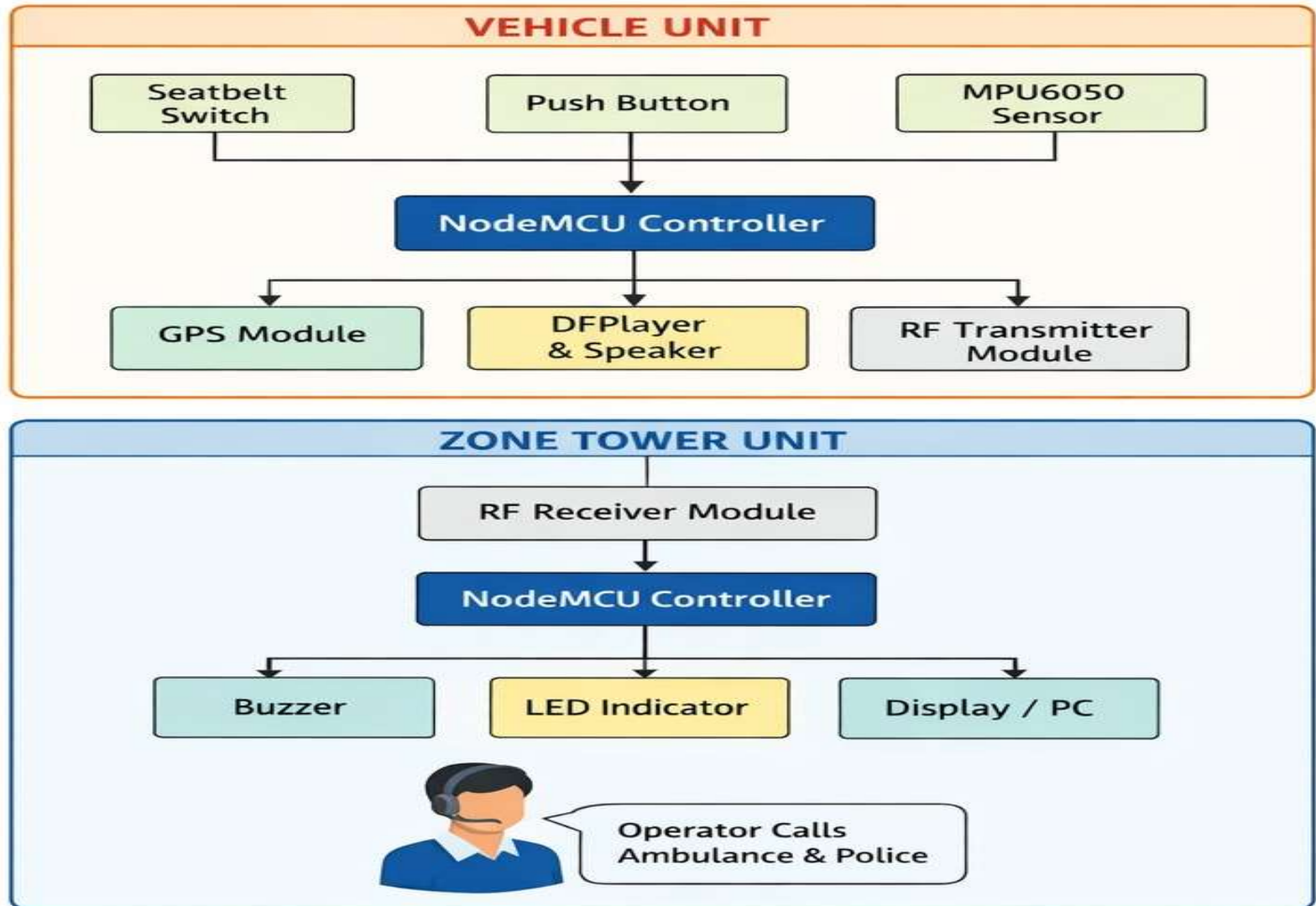
ALGORITHM

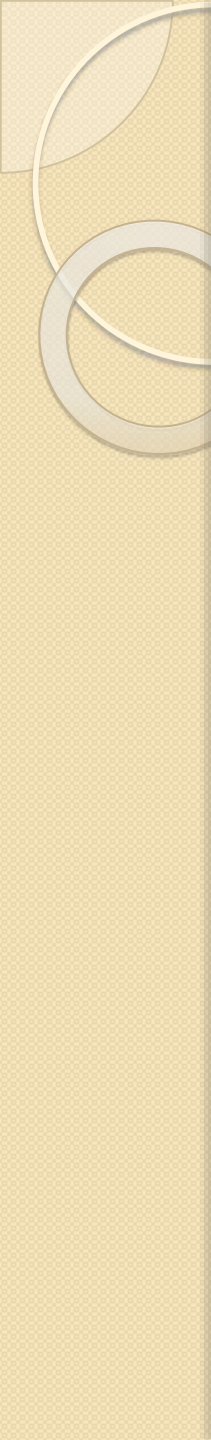
- 1) Start system
- 2) Check seatbelt status
- 3) Monitor vehicle motion using MPU6050
- 4) If abnormal impact detected
- 5) Play voice alert
- 6) Wait for driver response
- 7) If button pressed → cancel alert
- 8) Else capture GPS location
- 9) Send data through RF module
- 10) Tower alerts operator

FLOW DIAGRAM



BLOCK DIAGRAM





CONCLUSION

The proposed system provides an efficient solution for automatic accident detection and quick emergency communication. By reducing the time required to report accidents, it helps improve rescue response and can potentially save lives.

REFERENCES

Janani Sindhu P, Varshini K, Aarthi M, Fathima M, Hemavathi S, Kavitha V. AcciRadar: Real-Time Automatic Accident Detection and Reporting Framework. Proceedings of the 2025 International Conference on Computing and Communication Technologies (ICCCT), IEEE, 2025. DOI: 10.1109/ICCCT63501.2025.11019134.

Sumitra V, Bhargavi R, Tharani D, Heasreetha Reddy V, Veni N, Amrutha B. Collision Detection and Alert System for Road Safety. Proceedings of the 2025 International Conference on Computing and Communication Technologies (ICCCT), IEEE, 2025. DOI: 10.1109/ICCCT63501.2025.11019056.

M. Sumithra, B. Buvaneswari, Madhan S, Mohamed Shameer S, Pranith S. Vehicle Accident Prevention and Detection System Based on Mobile Application. Proceedings of the 2025 3rd International Conference on Communication, Security, and Artificial Intelligence (ICCSAI), IEEE, 2025. DOI: 10.1109/ICCSAI64074.2025.11064507.



THANK YOU